MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Elective (DSE) –
MG5DSEGEO300 – Geology and Geomorphology of Kerala
(2024 ADMISSION ONWARDS)
QP Code: 26900259

ANSWER SCHEME

PART A

Short answer in 60 words
Answer **any 5** questions
Each question carries **2 marks**

1. The lowland region of Kerala, locally known as Theeradesham (coastal region), is the narrow strip of land along the Arabian Sea. It extends along the western side of the state and is generally situated at an elevation of less than 7.5 metres above sea level. (**Definition-1 mark; Examples - 0.5 mark; Features-0.5 mark).**

2. Lateritization is the process of soil formation in hot and humid tropical regions through intense leaching of silica and other soluble minerals. It leads to the accumulation of iron and aluminium oxides in the soil, resulting in the formation of laterite soil. **(Definition- 2 marks).**

3. Supracrustal rocks are oldest rocks formed from sediments or volcanic materials deposited on the Earth's crust and later subjected to processes such as metamorphism. They commonly include metamorphosed sedimentary and volcanic rocks. **(Definition- 2 marks).**

4. Gneiss is a coarse-grained metamorphic rock formed under high temperature and pressure. It is characterised by alternating bands of light-coloured minerals, such as quartz and feldspar, and dark-coloured minerals, such as mica or amphibole. **(Definition- 2 marks).**

5. Basic intrusive rocks are dark-coloured igneous rocks formed by the slow cooling and solidification of magma beneath the Earth's surface. In Kerala, dolerite and gabbro are examples of basic intrusive rocks. **(Definition- 1 mark; Location in Kerala-1 mark).**

6. The major non-metallic mineral resources of Kerala include china clay (kaolin), silica sand, limestone, graphite, bauxite and mineral sands containing ilmenite, rutile and monazite. **(List of 6 - 2 marks).**

PART B

Short notes in 250 words
Answer **any 6** questions
Each question carries **8 marks**

7. Definition-2 marks; Formation & Georaphical features-2 marks; Major backwaters of Kerala-2 marks; Significance-2 marks.
8. Definition-2 marks; Origin-1 mark; Rock composition of different rocks in the Khondalite group-2 marks; Distribution in Kerala-2 marks; Geological significance-1 mark.
9. Definition of mud banks-2 marks; Process of formation of mud banks in Kerala-4 marks; Distribution in Kerala-2 marks.
10. Definition of granite plutons-1 mark; Names and geographical distribution of important granite plutons in Kerala-2 marks; Comparison (on mineral composition and textural characteristics, age, mode of emplacement and geological setting)-4 marks; Geological significance-1 mark.
11. Definition -2 marks; Major sources and distribution of sand mining in Kerala, including rivers and coastal areas-3 marks; Environmental consequences.-3 marks.
12. Introduction to metamorphic rocks-1 mark; Composition, characteristics, geological significance and occurrence in Kerala of Khondalite-2 marks; of Charnockite.-2 marks; of Gneiss-2 marks; of Leptynite-1 mark.
13. Definition-2 marks; four main of coastal regulation zones-4 marks; purpose of CZR-2 marks.
14. Introduction on Tertiary sedimentary sequence of Kerala-2 marks; Explanation on characteristics of Vaikom Formation-6 marks.

PART C

Essay in 400 words
Answer **any 1** questions
Each question carries **12 marks**

15. Definition of landslide-2 marks; Occurrence in Kerala-2 marks; Geological controls (factors)-2 marks; Geomorphological controls (factors)-2 marks; Triggering factors-2 marks ; Consequences-2 marks.
16. Definition and general characteristics of placer mineral deposits-2 marks; Origin-3 marks; Distribution in Kerala-2 marks; Mineral assemblage-2 marks; Economic importance-3 marks.